TravelFlow

System Requirements Specification

Course: Software Engineering Analysis & Design

Document: Functional & Non-Functional Requirements

This document specifies the complete functional and non-functional requirements for **TravelFlow** — a digital platform for bus ticket booking, vehicle rental, driver management, and transport fleet administration. Requirements are grounded in Albanian national legislation and international software engineering standards. Each functional requirement maps directly to a user story (US-01 through US-35) and a validated user scenario.

FR	Functional Requirement	NFR	Non-Functional Requirement

REQUIREMENTS

1 Functional Requirements

28 Requirements

Req #	Requirement	Description	Comments	SME Reviewed & Approved
FR-01	**User Registration**	The system shall allow new visitors to create an account by providing their full name, email address, and a password of at least 6 characters.	*Eliminates manual onboarding; auto-login on success for a seamless first impression.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-02	**Auto-Login After Registration**	Upon successful account creation, the system shall automatically authenticate the new user and redirect them to the platform.	*Removes friction — users land directly in the product, ready to explore.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-03	**User Login**	Registered users shall be able to log in using their email and password. The system shall create and persist a client-side session on success.	*Generic error messages ("Invalid email or password") protect against enumeration attacks.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-04	**Secure Logout & Session Expiry**	Authenticated users shall be able to log out via the navigation bar. Sessions shall expire automatically after a defined idle period.	*Prevents session hijacking on shared devices; all protected routes redirect to login post-logout.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-05	**Role-Based Access Control (RBAC)**	The system shall enforce three distinct roles — client, driver, admin — each with its own navigation, pages, and API permissions.	*Drivers cannot see admin pages; clients cannot reach the driver portal. Unauthorised URLs return Access Denied.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-06	**Trip Search Engine**	Clients shall be able to search for available trips by selecting departure city, destination, and travel date from a structured search panel.	*City pairs map to internal route IDs; identical From/To cities trigger a validation error before querying.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-07	**Automatic Trip Generation (autoTrips)**	If no trips exist in the database for the requested route and date, the system shall automatically generate the full day schedule based on predefined timetables.	*Guarantees zero-downtime service availability without manual admin intervention for every new date.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-08	**Trip Card Display**	Search results shall be displayed as cards showing route, departure time, arrival time, bus model, capacity, and base price.	*A clear "No trips found — try a different date" message guides users when results are empty.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-09	**Destination Stop Selection**	During booking, clients shall select their specific destination stop before proceeding; the ticket price shall update dynamically based on the chosen segment.	*Confirm Booking remains disabled until a stop is selected — preventing zero-price bookings.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-10	**Interactive Seat Map**	The booking flow shall present a visual seat layout showing occupied seats in red, free seats in green, and the selected seat highlighted in teal.	*Concurrent bookings of the same seat shall prompt the client to re-select; the map reflects live state.*	Etna Gjika Tristan Halo Dhimitri Shallo

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FR-11	**Itemised Fee Breakdown**	Before confirming, clients shall be shown a full breakdown: ticket price, site fee (100 lek), and seat selection fee (where applicable).	*Transparency requirement — no hidden charges; clients can expand the breakdown via "Show Fee Breakdown".*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-12	**Booking Confirmation & Unique Code**	On confirmation, the system shall generate a unique alphanumeric booking code, persist the Bookings record, and lock the seat in bookedSeats.	*The code is the passenger's travel credential and is displayed immediately via toast notification.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-13	**Random Seat Booking**	Clients shall have the option to request a randomly assigned seat at no extra charge, with the system selecting the lowest available seat number.	*Clients may switch to manual seat selection at any point during the booking flow.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-14	**Booking History & Digital Ticket**	Clients shall be able to view all their bookings in chronological order under My Account, and open any confirmed booking as a full digital ticket.	*Past trip tickets remain accessible for reference. Filter by status: All, Confirmed, Cancelled.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-15	**Booking Cancellation**	Clients shall be able to cancel a confirmed, non-departed booking via a two-step dialog. On cancel, the seat is released back to bookedSeats.	*The cancel button is hidden for already-cancelled or post-departure bookings — no accidental double-cancels.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-16	**Vehicle Rental Catalogue**	Clients shall be able to browse all available rental cars with make, model, year, fuel type, transmission, seat count, daily price, and a photo.	*Filter controls allow narrowing by category, fuel, and transmission. Unavailable cars (available=0) are hidden.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-17	**Rental Reservation with Price Confirmation**	Clients shall be able to select a car, choose pickup and return dates, and confirm a reservation with an automatically calculated total price.	*Validation prevents return dates on or before pickup date. Price = daily rate × number of rental days.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-18	**Rental History & Status Tracking**	Clients shall view all their rental reservations in My Account with car details, dates, total price, and current status (Confirmed / Completed).	*Empty state provides a direct link to browse the rental fleet.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-19	**Vehicle Return & Availability Reset**	Admins shall be able to mark a rental as returned; the system shall update the Rentals record to "completed" and set the car's available flag to 1.	*An automated daily process (checkRentals) also completes overdue confirmed rentals and frees the vehicle.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-20	**Driver Portal Access**	Users with role=driver shall see a dedicated portal on login showing their assigned trips with departure time, bus plate number, and passenger count.	*Non-driver roles navigating to the driver URL receive "Access Denied". No admin or client data is visible.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-21	**Ticket Validation**	Drivers shall be able to enter a passenger booking code and instantly receive a validation result: passenger name, seat, stop, and route.	*Duplicate/already-validated codes trigger a prominent "ALREADY VALIDATED" warning to deter fraud.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-22	**Continuous Boarding Validation**	The ticket validation interface shall support scanning multiple codes consecutively without a page refresh.	*Designed for speed at the bus door; input field auto-focuses after each validation result.*	Etna Gjika Tristan Halo Dhimitri Shallo

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FR-23	**Admin Dashboard with KPIs**	Admin users shall access a unified dashboard showing key metrics: total confirmed bookings, total revenue (lek), total users, total buses, and active rentals.	*All figures are read-only aggregates computed from live database counts — no risk of accidental modification.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-24	**Bus Fleet Management**	Admins shall be able to view and register buses with: Bus Code, Plate, Model, Capacity, Route, and Status. Duplicate codes/plates are rejected.	*Fleet visibility is essential for operational planning; inactive buses can be flagged before being assigned trips.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-25	**Rental Fleet Management**	Admins shall be able to add rental cars to the catalogue. Cars are immediately visible to clients once added with available=1.	*Admins can set available=0 to temporarily pull a car from the catalogue without deleting its record.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-26	**User Management Panel**	Admins shall view all registered users with their name, email, role, and registration date. Driver accounts can be created directly from this panel.	*New driver records include bus_id and route assignment so the driver portal populates correctly on first login.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-27	**Vehicle Maintenance Scheduler**	Admins shall view maintenance schedules for all vehicles, mark individual services as completed, and have the system auto-calculate the next due date.	*Each service type (oil, tyres, brakes, inspection, insurance, registration) has its own intervalDays and can be updated independently.*	Etna Gjika Tristan Halo Dhimitri Shallo
FR-28	**Rental Management & Monitoring**	Admins shall view all active and past rentals with client details, car info, dates, and status, and can process returns directly from this panel.	*Overdue rentals are highlighted; the checkRentals background job handles automatic completion as a safety net.*	Etna Gjika Tristan Halo Dhimitri Shallo

2 Non-Functional Requirements

6 Categories

Non-functional requirements specify criteria that govern the overall quality, performance, and compliance of TravelFlow — beyond individual feature behaviours. These requirements are mandatory for a robust, secure, and legally compliant system.

2.1 Security Requirements

- Encryption of all sensitive user data at rest and in transit (TLS 1.2+).
- Multi-Factor Authentication (MFA) mandatory for all administrative accounts.
- Role-Based Access Control (RBAC) enforced at both UI and API layers — no client can reach driver or admin endpoints.
- Periodic penetration testing and security audits at minimum every 6 months.
- Secure automated backup mechanisms with off-site storage to prevent data loss.
- Full compliance with GDPR and Albanian Law No. 124/2024 on Personal Data Protection.
- Generic authentication error messages to prevent credential enumeration.
- Booking codes generated with sufficient entropy to resist brute-force guessing.

2.2 Performance Requirements

- API response time shall be under 3 seconds for all standard operations under normal load.
- System availability (uptime) shall be at least 99.9% — particularly during peak travel booking periods.
- The platform shall support a minimum of 500 simultaneous authenticated users without performance degradation.
- The autoTrips generation process shall complete within 500ms so search results appear without noticeable delay.
- Database queries on the trips table (filtered by route and date) shall execute in under 200ms with proper indexing.

2.3 Reliability Requirements

- Automated database backups shall run daily with a maximum data loss window of 24 hours (RPO).
- Redundancy and failover mechanisms shall be in place for database availability.
- System recovery time after failure shall not exceed 15 minutes (RTO).
- Seat booking operations shall be atomic — concurrent requests for the same seat must be handled without double-booking.
- The checkRentals automated job shall run reliably daily to free overdue vehicle reservations.

2.4 Accessibility & Usability

- The user interface shall be intuitive and fully usable without prior training by any citizen.
- The platform shall be fully responsive and function correctly on desktop browsers, tablets, and mobile devices.
- All error messages shall be written in plain language and indicate clearly what the user should do next.
- The seat map colour scheme (red / green / teal) shall be accompanied by text labels to support colour-blind users.
- WCAG 2.1 Level AA compliance is the target accessibility standard for all client-facing pages.

2.5 Maintainability & Scalability

- The autoTrips schedule and pricing configuration shall be editable via a single data structure without code changes.
- The system architecture shall support horizontal scaling of the PHP API layer behind a load balancer.
- All API actions shall be logged with timestamp, user ID, and action type for audit trail purposes.
- Database schema changes shall follow versioned migration scripts to enable rollback.

2.6 Data Integrity & Compliance

- Every booking confirmation code shall be globally unique across the Bookings table (UNIQUE constraint).
- User passwords shall not be stored in plain text — the system shall implement secure hashing (bcrypt or equivalent).
- All personal data retention periods shall comply with Albanian data protection law and be documented in a Privacy Policy.

- Financial transaction data (booking totals, rental fees) shall be immutable after confirmation — no silent amendments.
- The statistics panel shall be strictly read-only; no write operations shall be possible from that interface.